E-DBPN: Enhanced Deep Back-Projection Networks for Remote Sensing Scene Image Superresolution

Yunlong Yu, Xianzhi Li, *Student Member, IEEE*, and Fuxian Liu

***Abstract*— Generative adversarial networks (GANs) have been widely used to single image superresolution (SR) (SISR), and these GAN-based methods have achieved significant performance on natural images due to their abilities to generate realistic textures. However, previous GAN-based methods are of poor performance when applied to remote sensing scene image SR. In order to further enhance the visual quality, in this article, we present a GAN-based SISR method by proposing a novel generator, which is capable of generating perceptually pleasing remote sensing scene images. First, we design the enhanced deep back-projection network (E-DBPN) generators based on the architecture of the original DBPN and mainly make two modifications. The first one is to add the proposed enhanced residual channel attention module (ERCAM) into the original DBPN, which can keep good properties of the original input features but also has the ability to emphasize more important features and suppressing less useful features. The other is to replace the concatenation operation with the proposed sequential feature fusion module (SFFM) for dealing with the feature maps generated by different up-projection units discriminatorily. As for the training process, the E-DBPN generator is first trained using the mean squared error (MSE) loss. Next, in order to improve the perceptual quality of the recovered images, we employ the content loss and the adversarial loss to train our initialized generator network. Experiments show that our method achieves state-of-the-art performance compared to other SISR methods.**

***Index Terms*— Attention mechanism, generative adversarial networks (GANs), perceptual quality, remote sensing scene image, single image superresolution (SISR).**

I. Introduction

NOWADAYS, the satellite imaging technologies are developed rapidly and provide many kinds of applications, such as land use and land cover scenes identification and classification [1], [2], environment monitoring [3], [4], and remote sensing image retrieval [5], [6]. In these applications, the demands for high-resolution (HR) images with high-frequency information never cease. Nevertheless, the quality of remote sensing scene images is influenced by many factors, e.g., the relative movement, the atmospheric interference, the ultra-distance imaging, and the channel transmission capacity. The spatial resolution of the remote sensing scene images can be whittled by these factors. Using advanced devices helps us obtain HR images, however, the deployment of advanced remote sensing instruments needs a high cost of launch and maintenance. Therefore, single image superresolution (SR) (SISR), i.e., recovering a single HR image from its low-resolution (LR) observed image has been a hot topic [7], [8].

In general, SISR is a very challenging and severely ill-posed inverse problem since there exist multiple HR images corresponding to a single LR image. In order to solve this inverse problem, plenty of conventional SR algorithms have been proposed for both natural images and remote sensing scene images, e.g., patch-based methods [9]–[12], prediction-based methods [13]–[17], edge-based methods [18], and sparse representation methods [19]–[21]. However, these conventional SR methods cannot perform well due to the insufficient additional information and the restricted representation capacity of their used models.

In more recent years, convolutional neural network (CNN), a crucial branch of the methods of deep learning, has been attracting more research attention due to its amazing performance in various computer vision tasks, such as image segmentation [22], [23], image classification [24], [25], and image retrieval [26], [27]. In particular, CNN-based SISR methods have achieved state-of-the-art performance on various publicly available SISR benchmark data sets, which shows great superiority over conventional SR methods. Therefore, in this article, we only focus on the deep learning-based SISR methods. According to the SR framework, the deep learning-based SISR methods can be roughly divided into four categories, i.e., pre-upsampling SR framework, post-upsampling SR framework, progressive upsampling SR framework, and iterative up and down sampling SR framework. In these deep learning-based SISR methods, most of the methods are proposed for natural images, only a few methods are proposed for remote sensing scene images [28]–[34]. Therefore, there is still room for further improving the performance at the SISR task of remote sensing scene images. The difference between natural image SR and remote sensing scene image SR is the input object. Compared with natural images, remote sensing scene images have complex degradation process. The quality of remote

Manuscript received October 8, 2019; revised November 24, 2019 and December 28, 2019; accepted January 11, 2020. Date of publication January 29, 2020; date of current version July 22, 2020. This work was supported by the National Natural Science Foundation of China under Grant 71771216 and Grant 71701209. *(Corresponding author: Xianzhi Li.)*

Yunlong Yu and Fuxian Liu are with the Air Defense and Anti-Missile College, Air Force Engineering University, Xi'an 710051, China.

Xianzhi Li is with The Chinese University of Hong Kong, Hong Kong (e-mail: lixianzhi123@gmail.com).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2966669

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

sensing scene images could be affected by many additional factors, e.g., atmospheric disturbance, ultralong range imaging, and relative motion. Furthermore, compared with natural images, remote sensing scene images have a complicated background, low ground object resolution, and weak texture. Thus, there are considerable challenges in the field of remote sensing scene image SR. In the process of designing a SISR network for remote sensing scene images, we should pay more attention to how to finely process different types of features. Existing remote sensing scene image SR methods all treat different types of features equally, without rescaling different types of features adaptively. Hence, the performance of these methods is not satisfactory.

In this article, we present a generative adversarial network (GAN)-based SISR method by proposing a novel generator, which can generate perceptually pleasing remote sensing scene images. First, we design the enhanced deep backprojection networks (E-DBPN) generator on the basis of the architecture of the original DBPN [35] and mainly make two modifications for further improving the recovered image quality of DBPN. The one is to add the proposed enhanced residual channel attention module (ERCAM) into DBPN, which facilitates to keep good properties of the original input features, but also has the ability to emphasize more important features and suppressing less useful features. In the ERCAM, we employ max-pooling, L_p-pooling, and average-pooling to squeeze the spatial information of the input feature map, respectively, different from previous approaches. The other is to replace the concatenation operation with the proposed sequential feature fusion module (SFFM) for dealing with the feature maps generated by different up-projection units discriminatorily. With regard to the loss functions, the E-DBPN generator is first trained using the mean squared error (MSE) loss. Next, in order to improve the perceptual quality of the recovered images, we employ the content loss and the adversarial loss to train our initialized generator network. Experimental results demonstrate that our proposed GAN-based SISR method obtains better performance when compared with state-of-the-art SISR methods.

Overall, the contribution of this article is summarized in the following.

1) We design an ERCAM. In ERCAM, in order to aggregate sufficient information in each channel, the max-pooling, L_p-pooling and average-pooling are employed simultaneously to squeeze the spatial information of the input feature map. This module is capable of keeping good properties of the original input features, but also has the ability to emphasize more important features and suppressing less useful features.
2) We design an SFFM. In SFFM, the feature maps are integrated in a sequential manner. This module has the ability to discriminate in that the feature maps generated by different up-projection units are processed in different depths of the network.
3) We evaluate the effectiveness of our method on five widely used data sets, i.e., the NWPU-RESISC45 data set, the AID data set, the UC-Merced data set, the WHU-RS19 data set, and the RSSCN7 data set. The results show that our method outperforms the state-of-the-art SISR methods.

The rest of this article is organized as follows. In Section II, the related works are briefly reviewed. Section III presents the detailed description of our method. Section IV gives the experimental settings and results. Finally, Section V summarizes this article and presents our idea for future work.

II. Related Works

A. Frameworks of Existing Superresolution Models

Recently, the topic of SISR has been widely explored due to its extensive applications. Here, we will focus on deep learning based methods. A variety of deep learning based SR models have been proposed with impressive performance. These models are trained in a supervised manner, in other words, both LR images and the corresponding groundtruth HR images are involved. Here, we follow [35] to divide the frameworks of existing SR models into four groups: pre-upsampling SR framework, post-upsampling SR framework, progressive upsampling SR framework, and iterative up and down sampling SR framework. In what follows, we briefly review each category of the approaches.

1) Pre-upsampling Superresolution Framework: Preupsampling SR framework usually employs traditional predefined upsampling methods, e.g., bicubic interpolation, to produce coarse HR images and then using deep CNNs to obtain higher quality details from the coarse HR images. As a pioneer work, Dong *et al.* [36], [37] proposed the SR CNN (SRCNN) to learn the mapping from LR to HR images in an end-to-end manner. Subsequently, this framework has been one of the most popular frameworks in the field of SISR [38]–[44]. In the field of remote sensing scene image SR, most of the methods employ this framework. Lei *et al.* [29] proposed a local–global combined networks (LGCNet) for capturing restoration features at both local and global levels in the spatial domain. Wang *et al.* [32] developed a remote sensing image SR method by training CNNs with respect to wavelet analysis. Ma *et al.* [28] proposed a method to achieve remote sensing image SR via the wavelet transform combined with the recursive Res-Net. However, this framework may produce noise from the coarse HR images. Furthermore, time and space consumption is very high because most operations are conducted in high dimension.

2) Post-upsampling Superresolution Framework: For the sake of resolving the issues in pre-upsampling SR framework, post-upsampling SR framework is designed, in which a large proportion of mapping operations are conducted in low dimension. The upsampling layer is located at the end of the model. It is worth noting that the upsampling layer in this framework is learnable instead of the predefined one in pre-upsampling SR framework. This framework is first proposed by efficient subpixel CNN (ESPCN) [45] and fast SRCNNs (FSRCNN) [46]. On account of the reduction of the computational complexity, this framework has also been widely employed by most of researchers [47]–[54]. In the field of remote sensing scene image SR, deep distillation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

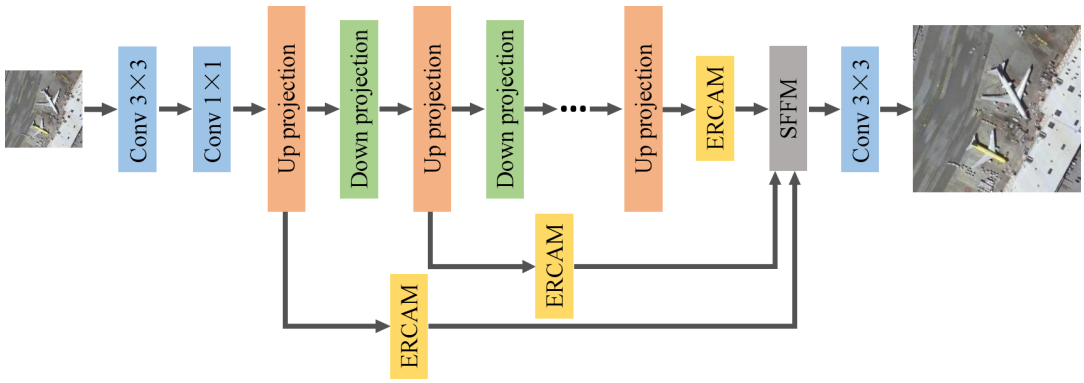


Fig. 1. Architecture of the E-DBPN generator.

recursive network [31] is designed, which includes a group of ultradense residual blocks, a multiscale purification unit, and a reconstruction module. Nevertheless, this framework cannot deal with the task of multiscale SR.

3) Progressive Upsampling Superresolution Framework: Due to the fact that the post-upsampling SR framework cannot provide a good solution for the multiscale SR task, a progressive upsampling SR framework is proposed in the Laplacian pyramid SR network (LapSRN) [55]. Subsequently, the multiscale LapSRN (MS-LapSRN) [56] and the progressive SR (ProSR) [57] have also employed this type of framework. This framework can be seen as a cascade of the post-upsampling SR framework. It progressively obtains the multiple-scale SR images with different scaling factors simultaneously. However, it still has some issues, e.g., the complexity of training.

4) Iterative Upsampling and Downsampling Superresolution Framework: In consideration of the reconstruction error in each stage, the iterative upsampling and downsampling SR framework is designed in DBPNs [35]. This framework uses multiple up- and down-projection units and provides the error feedback mechanism for refining the results at each stage, leading to relatively high performance.

Given that this framework has the advantage of capturing the deep relationships between the LR and the corresponding HR images, it has been a promising framework in the field of SISR [58]. In our article, we also adopt this framework as our foundation and make some improvements for the purpose of obtaining higher quality SR images.

B. Loss Functions

In this field, loss functions play an important role in optimizing the proposed model. Loss functions help us estimate the difference between the recovered SR images and the corresponding groundtruth HR images. In this section, we briefly review several loss functions commonly used for SISR.

1) Pixel Loss: This loss function evaluates the pixel-wise difference between two images and has a close relationship with the definition of peak signal-to-noise ratio (PSNR), which means that minimizing the pixel loss is equal to maximizing the PSNR. The purpose of this loss function is to let the recovered SR images be closer to the corresponding groundtruth HR images in terms of the pixel value. Pixel loss usually contains L1 loss and L2 loss. Nevertheless, this loss function may generate perceptually unsatisfied images because it does not take the quality of the image into consideration [59], [60].

2) Content Loss: Unlike the pixel loss, the content loss pays more attention to the image perceptual quality. In this loss, the Euclidean distance between two high-level representations extracted by the pretrained CNNs [61] is calculated to estimate the semantic difference between the two images. By employing this loss function, we can get the recovered SR images which are very similar to the corresponding groundtruth HR images in terms of perceptual quality. At present, most of researchers adopt this loss function in the field of SISR [47], [50], [53], [62]–[64].

3) Adversarial Loss: Recently, the GANs [65] have been shown to excel in many visual applications. GANs consist of a generator and a discriminator that compete in a two-player minimax game. The discriminator tries to distinguish real training data from synthetic data, and the generator tries to produce fake samples and fool the discriminator. In the field of SISR, the adversarial loss function has been widely used [47], [50], [53], [54]. The SR framework is regarded as the generator, while the discriminator tries to judge whether the input image is the recovered SR image or the groundtruth HR image.

4) Other Loss Functions: In addition to the loss functions mentioned above, some other loss functions, i.e., texture loss [66], [67], cycle consistency loss [68], total variation loss [69], and prior-based loss [70], are also utilized in the SR field. In practical application, multiple loss functions are commonly combined in one model for the balance of various aspects.

III. Method

A. Overview

In this article, the goal is to achieve SR remote sensing scene image I^{SR} from a given LR image I^{LR}, at the same time, the superresolved image I^{SR} should be very similar to the original HR remote sensing scene image I^{HR}. With regard to an image with L color channels, we describe I^{LR} by a real-valued tensor of size $M' \times N' \times L$ and I^{HR}, I^{SR} by $M \times N \times L$, respectively, where $M = rM'$, $N = rN'$, and r is the scaling factor.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

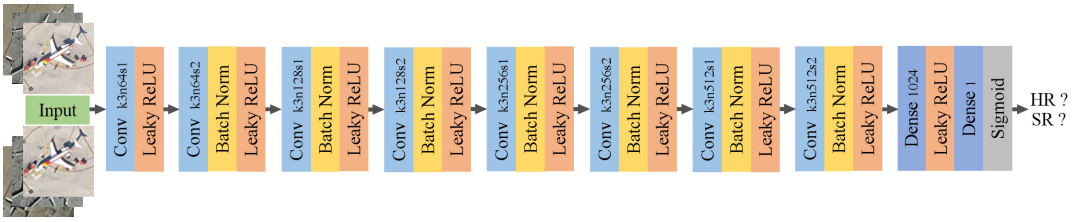


Fig. 2. Architecture of the utilized discriminator network.

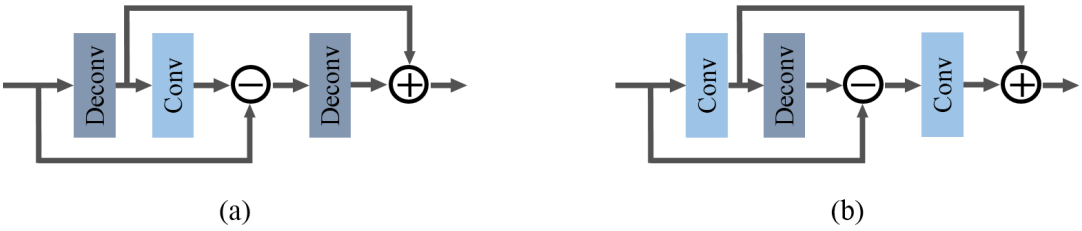


Fig. 3. Architectures of up- and down-projection units. (a) Architecture of the up-projection unit. (b) Architecture of the down-projection unit.

In order to obtain SR remote sensing scene image I^{SR} from a given LR image I^{LR}, we design the E-DBPN generators, which is shown in Fig. 1. We design this generator on the basis of the architecture of the DBPN [35] and mainly make two modifications for further improving the recovered image quality of DBPN: 1) add the proposed ERCAM into DBPN and 2) replace the concatenation operation with the proposed sequential feature fusion operation.

With regard to the loss functions, the E-DBPN generator is first trained using the pixel loss. However, this loss function may lead to producing a blurry and perceptually unsatisfying image. In order to improve the perceptual quality of I^{SR}, we employ the content loss and the GAN-based loss functions. We use the same discriminator network proposed in [47], which is shown in Fig. 2.

In the following, we first introduce the projection units in Section III-B. Then, we present the ERCAM in Section III-C. Next, we show how to conduct sequential feature fusion in Section III-D. Lastly, we explain the training loss functions in detail in Section III-E.

B. Projection Units

The projection units in our proposed framework include the up-projection unit and the down-projection unit. These two projection units are first designed in DBPN [35]. In what follows, we briefly describe these two projection units.

The up-projection unit is utilized to map the LR feature maps to the HR feature maps, which is shown in Fig. 3(a). This unit takes the LR feature maps as input and put them into high resolution through a deconvolutional layer. Then, a convolutional layer is employed to map the intermediate HR feature maps back to the LR feature maps. Next, the differences between the input LR feature maps and the reconstructed LR feature maps are put into high resolution through another deconvolutional layer. Finally, the output HR feature maps are obtained by adding the outputs of two deconvolutional layers.

The downprojection unit is conducted in a similar way, which is illustrated in Fig. 3(b). However, the purpose of the downprojection unit is to map the HR feature maps to the LR feature maps.

C. Enhanced Residual Channel Attention Module

The LR remote sensing scene images contain abundant low-frequency and valuable high-frequency features. The low-frequency features seem to be more complanate. The high-frequency features would usually be regions, being full of edges, texture, and other details. In the field of SISR, high-frequency channelwise features are more important for the HR reconstruction. If we treat the low-frequency and high-frequency features equally across channels, the representation capability of CNNs will be hindered. Therefore, we employ the channel attention mechanism to adaptively rescale each channelwise feature.

It is a common sense that our human beings tend to focus on some salient areas after conducting a sequence of partial glimpses [71]. Inspired by the human attention mechanism, various attempts have been made to introduce such attention mechanism into CNNs [72]–[74]. Their experimental results demonstrate that the attention mechanism can not only tell what is informative but also further boost the feature extraction ability. Our goal is to let the network have the ability to obtain informative and significant feature representations by designing an attention module. In other words, our network should pay more attention to important features and put down nonsignificant ones. We borrow ideas from squeeze-and-excitation (SE) block [74] and residual channel attention network (RCAN) [52], and make some improvements. The proposed attention module used in our framework is named as ERCAM, which are shown in Fig. 4. The ERCAM is capable of keeping good properties of the original input features but also has the ability to emphasize more important features and suppressing less useful features. The whole procedure of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

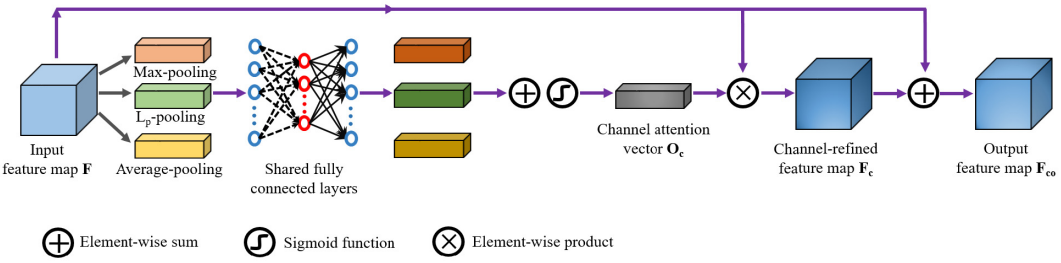


Fig. 4. Architecture of the ERCAM.

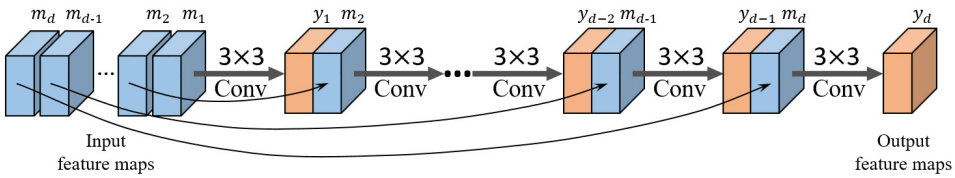


Fig. 5. Architecture of the SFFM. Note that each input feature map marked by light blue cube represents the output of associated ERCAM.

ERCAM is as follows.

$$\mathbf{F_c} = \mathbf{O_c} \otimes \mathbf{F} \tag{1}$$

$$\mathbf{F_{co}} = \mathbf{F_c} + \mathbf{F} \tag{2}$$

where $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ denotes the intermediate feature map, $\mathbf{O_c}$ represents the channel attention vector, $\mathbf{F_{co}}$ is the output feature map, and $\otimes$ represents element-wise multiplication. The following part gives the details of the module.

To aggregate information in each channel, the max-pooling and average-pooling operations are commonly employed. Additionally, L_p-pooling [75] is another choice to gather other distinctive clues, which can be regarded as a tradeoff between the max-pooling and average-pooling. Considering that different pooling operations may gather different important clues, in our work, we employ all the three pooling operations simultaneously to squeeze the spatial information of the input feature map. After that, we obtain three channel vectors, where the size of each channel vector is $\mathbb{R}^{C\times 1\times 1}$. We then feed each channel vector into the shared fully connected layers (SFCL) with one hidden layer. Note that, the number of hidden units U is less than the input dimension C. After employing the elementwise summation followed by an activation function, we can get the channel attention vector $\mathbf{O_c} \in \mathbb{R}^{C\times 1\times 1}$. Hence, we compute the channel-refined feature map $\mathbf{F_c}$ as follows:

$$\begin{aligned}\mathbf{F_c} = \sigma\,&(\mathrm{SFCL}(\mathrm{MaxPool}_s(\mathbf{F})) + \mathrm{SFCL}(L_p\mathrm{Pool}_s(\mathbf{F}))\\ &+ \mathrm{SFCL}(\mathrm{AvgPool}_s(\mathbf{F}))) \otimes \mathbf{F}\end{aligned} \tag{3}$$

$$\begin{aligned}&\mathrm{SFCL}(\mathrm{MaxPool}_s(\mathbf{F}))\\ &\quad = \mathrm{W}_2(\delta(\mathrm{W}_1(\mathrm{MaxPool}_s(\mathbf{F})) + b_1)) + b_2\end{aligned} \tag{4}$$

$$\begin{aligned}&\mathrm{SFCL}(L_p\mathrm{Pool}_s(\mathbf{F}))\\ &\quad = \mathrm{W}_2(\delta(\mathrm{W}_1(L_p\mathrm{Pool}_s(\mathbf{F})) + b_1)) + b_2\end{aligned} \tag{5}$$

$$\begin{aligned}&\mathrm{SFCL}(\mathrm{AvgPool}_s(\mathbf{F}))\\ &\quad = \mathrm{W}_2(\delta(\mathrm{W}_1(\mathrm{AvgPool}_s(\mathbf{F})) + b_1)) + b_2\end{aligned} \tag{6}$$

where δ denotes the rectified linear units (ReLU) function, σ denotes the sigmoid function, $\mathrm{W}_1 \in \mathbb{R}^{U\times C}$ and $\mathrm{W}_2 \in \mathbb{R}^{C\times U}$ are the weights in SFCL, $\mathrm{b}_1 \in \mathbb{R}^U$ and $\mathrm{b}_2 \in \mathbb{R}^C$ are the biases in SFCL. Finally, the channel-refined feature map $\mathbf{F_c}$ and the input feature map $\mathbf{F}$ are combined by employing element-wise summation, which results in the final output feature map $\mathbf{F_{co}}$.

D. Sequential Feature Fusion Module

In the DBPN model, the feature maps produced in each up-projection unit are concatenated for the final reconstruction. However, this operation does not have the ability to deal with the feature maps generated by different up-projection units discriminatorily. In general, the feature maps generated by different up-projection units have different types of the HR components and different impacts on the quality of the results [35]. Thus, it is necessary to deal with these feature maps discriminatorily. In our framework, we propose to integrate these feature maps in a sequential manner. Fig. 5 shows the illustration of the SFFM. Suppose that m_k represents the kth input feature map, y_k denotes the output of the kth convolutional layer. Next, we obtain the following equation:

$$y_k = \mathrm{conv}([m_k; y_{k-1}]) \tag{7}$$

where $k = 1, 2, \ldots, d$, $y_0 = 0$. d denotes the number of up-projection units, $[\cdot\,;\cdot]$ represents the concatenation operation and conv denotes the convolution operation. It is worth pointing out that the proposed SFFM has the ability to discriminate in that the feature maps generated by different up-projection units are processed in different depths of the network.

E. Loss Function

The training process of our proposed framework includes two steps: 1) initial training and 2) adversarial training. In the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

step of initial training, we employ the MSE loss to train our E-DBPN generator

$$L_{\mathrm{MSE}} = \frac{1}{\mathrm{MNL}} \sum_{i,j,q} \left(I^{\mathrm{HR}}_{i,j,q} - I^{\mathrm{SR}}_{i,j,q} \right)^2. \tag{8}$$

Utilizing the MSE loss helps us obtain the recovered images with high PSNR. Nevertheless, it may neglect high-frequency information and generate a perceptually unsatisfying image.

To remedy this, we add the step of adversarial training to train our initialized generator network. In this step, we employ the content loss and the adversarial loss to improve the perceptual quality, i.e., minimizing the semantic differences between images.

With regard to the content loss, we take pre-trained VGG-19 image classification network [61] as our feature extractor, in which the pretrained network is obtained using the ImageNet data set [76]. First, the feature extractor is utilized to obtain high-level representations on the v-th layer from I^{HR} and I^{SR}. Then, the L2 loss between the two extracted feature maps is defined as the content loss L_{content}

$$L_{\mathrm{content}} = \frac{1}{M_v N_v L_v} \sum_{i,j,q} \left(\phi^v_{i,j,q}(I^{\mathrm{HR}}) - \phi^v_{i,j,q}(I^{\mathrm{SR}}) \right)^2 \tag{9}$$

where M_v, N_v, and L_v represent the dimensions of the extracted feature map on the vth layer, $\phi^v(\cdot)$ denotes the representations on the vth layer.

The adversarial loss, i.e., GAN loss, is defined as follows:

$$L_a = -\log(d(I^{\mathrm{SR}})) \tag{10}$$

$$L_d = -\log(d(I^{\mathrm{HR}})) - \log(1 - d(I^{\mathrm{SR}})) \tag{11}$$

where $d(\cdot)$ represents the output of the discriminator, L_a is the GAN loss for the generator, and L_d is the GAN loss for the discriminator. Finally, the total loss of the generator L_g is

$$L_g = L_{\mathrm{content}} + \eta L_a \tag{12}$$

where η is the weighting parameter for the GAN loss.

IV. Experiments

In this section, we first introduce the data sets used for the experiments and the details about the training process. And then, we perform extensive ablation experiments to thoroughly evaluate the effectiveness of our design choice. Next, we assess the performance of the initialized generator network. Finally, the GAN-based final generator network is evaluated.

A. Data Set Collection

As for natural image SR, a large number of training and evaluating data sets, e.g., ImageNet [76], DIV2K [77], Set5 [78], Set14 [79], and BSD300 [80], are available. Nevertheless, there are no open data sets for SR of remote sensing scene images. Therefore, we collect some remote sensing scene classification data sets with relatively high spatial resolution and make full use of them to evaluate our method. In the following part, we first introduce these data sets. Then, we present how to utilize these data sets to train and evaluate our method.

1) *UC-Merced Data Set [81]:* This data set consists of 2100 overhead scene images with 256 × 256 pixels evenly from 21 classes in the RGB color space. These images are selected from the United States Geological Survey (USGS) National Map.
2) *WHU-RS19 Data Set [82]:* This data set is extracted from a set of satellite images exported from Google Earth with various pixel resolutions up to half a meter. It contains 1 005 images with 600 × 600 pixels from 19 categories.
3) *RSSCN7 Data Set [83]:* This data set is collected from Google Earth. It contains 2 800 aerial scene images with 400 × 400 pixels from seven classes. There are 400 images in each category.
4) *AID Data Set [84]:* This data set is collected from Google Earth, which contains 10 000 images from 30 classes with 600 × 600 pixels. The class size varies from 220 to 420. Note that images in this data set are captured from different countries in different seasons.
5) *NWPU-RESISC45 Data Set [85]:* This data set is a large-scale remote sensing scene data set, which contains 31,500 images with 256 × 256 pixels evenly from 45 scene classes. The spatial resolution varies from about 30 to 0.2 m per pixel. The characteristics of this data set include large scale, high within-class diversity, high between-class similarity, and rich image variations.

In the process of initial training, we employ the NWPU-RESISC45 data set to train and evaluate the E-DBPN generators. We randomly sample 90 images from this data set for evaluation, and the remaining for training. In order to expand the volume of training data, we use a moderate data augmentation through flipping them horizontally and vertically and rotating them by 90°. Furthermore, we also use 105 images randomly sampled from the RSSCN7 data set to evaluate our method. These two evaluation sets are named Set90 and Set105, respectively.

With regard to training the GAN-based final generator, we use the AID data set. Data augmentation is also implemented in this data set. For evaluation, we use the UC-Merced data set and the WHU-RS19 data set.

B. Implementation Details

In our experiments, we use a workstation equipped with the Intel Xeon(R) E5-1620 processor with 10 M of Cache and up to 3.6 GHz, 64-GB memory, a NVIDIA GeForce GTX1080Ti GPU with a 11-GB memory, and a SAMSUNG SSD 850 PRO with 1 TB of capacity. Our method is implemented using TensorFlow.

In our experiments, we mainly focus on the upscaling factor of 4 between LR and HR images, which is usually the most challenging case in SISR [31], [47], [54]. The HR images are obtained from the given data sets, in which all images are resized to 256 × 256 pixels. Then, these images are downsampled by employing the bicubic interpolation, which can obtain the LR images. The intensity ranges of HR and LR images are normalized to $[-1, 1]$. The weighting parameter for the GAN loss η is set to 10^{-3} . With regard to the content

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT SISR METHODS ON THE SET90 DATA SET (2× UPSCALING)

Method	Evaluation criteria	
	PSNR	SSIM
Bicubic	31.86	0.9025
SRCNN [37]	34.18	0.9418
FSRCNN [46]	34.40	0.9442
VDSR [38]	34.82	0.9484
LapSRN [55]	34.86	0.9489
LGCNet [29]	34.88	0.9492
WTCRR [28]	35.16	0.9535
D-DBPN [35]	35.61	0.9546
RCAN [52]	35.75	0.9573
E-DBPN$_{\mathrm{M}}$ (ours)	**35.86**	**0.9586**

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT SISR METHODS ON THE SET105 DATA SET (2× UPSCALING)

Method	Evaluation criteria	
	PSNR	SSIM
Bicubic	30.85	0.8791
SRCNN [37]	32.72	0.9258
FSRCNN [46]	32.90	0.9301
LapSRN [55]	33.24	0.9332
VDSR [38]	33.29	0.9342
LGCNet [29]	33.34	0.9349
WTCRR [28]	33.61	0.9372
D-DBPN [35]	33.67	0.9384
RCAN [52]	33.82	0.9412
E-DBPN$_{\mathrm{M}}$ (ours)	**33.89**	**0.9422**

loss, we employ the Conv5 layer in VGG-19 since this layer can achieve better results than other layers.

The process of training is divided into two steps, i.e., initial training and adversarial training. First, we train the E-DBPN with the MSE loss. We conduct about 10^6 update iterations with an initial learning rate of 10^{-4}. The learning rate is decreased by $1/10$ when the training loss stops decreasing. When the learning rate is equal to 10^{-6}, the learning rate will not be decreased. Next, the trained MSE-based E-DBPN, termed as E-DBPN$_{\mathrm{M}}$, is employed as an initialization for the generator when training the actual GANs, named as E-DBPN$_{\mathrm{G}}$. Initial training with MSE loss can help the GAN-based model avoid undesired local optima. As for the adversarial training, we perform about 2×10^5 update iterations. We use 10^{-4} as the learning rate for the first 10^5 iterations, 10^{-5} for the next 10^5 iterations. For optimization, we use Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$.

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT SISR METHODS ON THE SET90 DATA SET (4× UPSCALING)

Method	Evaluation criteria	
	PSNR	SSIM
Bicubic	27.60	0.7507
SRCNN [37]	29.15	0.8012
FSRCNN [46]	29.28	0.8056
VDSR [38]	29.73	0.8201
LGCNet [29]	29.76	0.8217
LapSRN [55]	29.92	0.8247
WTCRR [28]	30.10	0.8256
SRResNet [47]	30.29	0.8336
SRFeat$_{\mathrm{M}}$ [54]	30.48	0.8370
D-DBPN [35]	30.59	0.8397
RCAN [52]	30.65	0.8429
E-DBPN$_{\mathrm{M}}$ (ours)	**30.86**	**0.8553**

C. Evaluation of the Initialized Generator

In this section, we employ two classical evaluation criteria, i.e., PSNR [dB] and structural similarity index measure (SSIM) to evaluate the performance of the initialized generator, i.e., E-DBPN$_{\mathrm{M}}$, and make a comparison with ten state-of-the-art methods: SRCNN [37], fast SRCNNs (FSRCNN) [46], VDSR [38], local–global combined networks (LGCNets) [29], wavelet transform combined with the recursive Res-Net (WTCRR) [28], Laplacian pyramid SR network (LapSRN) [55], SR residual network (SRResNet) [47], SRFeat$_{\mathrm{M}}$ [54], dense DBPNs (D-DBPN) [35], RCAN [52].

Tables I–IV show the quantitative comparisons for $\times 2$, $\times 4$ SR on the Set90 data set and the Set105 data set, respectively. Note that the best values are marked in bold. We take the $\times 4$ SR, for example. From these tables on scale $\times 4$ we can see that, overall, our method, i.e., E-DBPN$_{\mathrm{M}}$, could achieve the highest performance on all data sets compared with others. Specifically, on the Set90 data set (4× upscaling), E-DBPN$_{\mathrm{M}}$ boosts the second-best method, i.e., RCAN, by 0.21 dB in PSNR, 0.0124 in SSIM. On the Set105 data set (4× upscaling), the PSNR gain of E-DBPN$_{\mathrm{M}}$ over RCAN is 0.16 dB, and the SSIM gain of E-DBPN$_{\mathrm{M}}$ over RCAN is 0.0112. The good performance of our method benefits from the proposed ERCAM and SFFM.

In addition to quantitative evaluations, visual comparisons are also made between the proposed method and the other methods. In Fig. 6, we show the visual comparisons. From the recovered images of different methods in Fig. 6, we can see that our proposed method has a stronger ability to alleviate the blurring artifacts and recovering more details, which is more faithful to the groundtruth.

Furthermore, we also compare variations of our method to see the effect of each major element. In this section, we empirically analyze the major elements in our model by removing or replacing each of them when we train our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Visual comparisons for 4× SR between our model with the MSE loss (E-DBPN$_M$) and other methods. (a) HR. (b) LR. (c) Bicubic. (d) D-DBPN. (e) RCAN. (f) E-DBPN$_M$ (ours).

model: 1) *w/o ERCAM*—we remove the ERCAM from our model; 2) *RCAB*—we replace the ERCAM with the residual channel attention block (RCAB) [52] in our model; and 3) *Concatenation Operation*—we replace the SFFM with the concatenation operation in our model. Tables V and VI show the PSNR and SSIM of employing three different settings on the Set90 data set and the Set105 data set, respectively. From the results in Tables V and VI, we can observe that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT SISR METHODS ON THE SET105 DATA SET (4× UPSCALING)

Method	Evaluation criteria	
	PSNR	SSIM
Bicubic	27.42	0.7297
SRCNN [37]	28.40	0.7743
FSRCNN [46]	28.50	0.7806
VDSR [38]	28.82	0.7913
LGCNet [29]	28.84	0.7929
LapSRN [55]	28.85	0.7940
WTCRR [28]	29.08	0.7972
SRResNet [47]	29.13	0.8029
SRFeat_{M} [54]	29.19	0.8061
D-DBPN [35]	29.28	0.8089
RCAN [52]	29.33	0.8129
E-DBPN$_{\text{M}}$ (ours)	**29.49**	**0.8241**

TABLE V
QUANTITATIVE COMPARISON OF VARIATIONS OF OUR GENERATOR NETWORK BY EMPLOYING DIFFERENT SETTINGS ON THE SET90 DATA SET (4× UPSCALING)

Different settings	Evaluation criteria	
	PSNR	SSIM
w/o ERCAM	30.71	0.8452
RCAB	30.76	0.8497
Concatenation Operation	30.79	0.8528
Our full pipeline	30.86	0.8553

TABLE VI
QUANTITATIVE COMPARISON OF VARIATIONS OF OUR GENERATOR NETWORK BY EMPLOYING DIFFERENT SETTINGS ON THE SET105 DATA SET (4× UPSCALING)

Different settings	Evaluation criteria	
	PSNR	SSIM
w/o ERCAM	29.36	0.8141
RCAB	29.39	0.8177
Concatenation Operation	29.41	0.8213
Our full pipeline	29.49	0.8241

the PSNR/SSIM of our full pipeline, i.e., E-DBPN$_{\text{M}}$, achieve 30.86/0.8553 and 29.49/0.8241 on the Set90 data set and the Set105 data set, respectively, which outperform all the other settings. The good performance of our full pipeline verifies the effectiveness of the framework elements being explored here.

D. Evaluation of the GAN-Based Final Generator

In this section, we will evaluate the performance of the GAN-based final generator in a different way. The purpose of employing the content loss and the GAN-based loss functions is to improve the perceptual quality of the superresolved images. However, existing evaluation criteria, e.g., PSNR and SSIM, are not suitable to estimate the perceptual quality of the superresolved images. As we all know, the SISR methods can serve as the preprocessing step for improving the performance of other computer vision tasks, e.g., face recognition [86]. Therefore, Sajjadi *et al.* [50] proposed to employ the object recognition performance as the metric to assess the image reconstruction methods, which is similar to human perception. In their article, they first downsample the original HR images in the validation data set. And then, they use the SISR method to obtain the superresolved images from the downsampled ones. Finally, these superresolved images are fed into the pretrained object recognition model obtained using the ImageNet data set [76]. In this way, we can obtain the classification accuracy. At the same time, the original HR images in the validation data set are also fed into this recognition model, in which its classification accuracy serves as baseline. They assume that higher classification accuracy implies that the superresolved images are closer to the original HR images. In this section, we also adopt this evaluated metric to assess the perceptual quality of the superresolved images produced by our method and other SISR methods.

TABLE VII
PERFORMANCES OF CLASSIFICATION TESTS USING IMAGES FROM THE TESTING SET OF THE UC-MERCED DATA SET (4× UPSCALING)

Method	Training ratios (%)	
	50%	80%
Bicubic	64.57%	66.43%
E-DBPN$_{\text{M}}$	79.62%	81.19%
$\text{SRFeat}_{\text{IF}}$ [54]	86.86%	88.10%
E-DBPN$_{\text{G}}$	90.19%	91.43%
Baseline (HR)	94.10%	95.24%

TABLE VIII
PERFORMANCES OF CLASSIFICATION TESTS USING IMAGES FROM THE TESTING SET OF THE WHU-RS19 DATA SET (4× UPSCALING)

Method	Training ratios (%)	
	40%	60%
Bicubic	65.96%	67.11%
E-DBPN$_{\text{M}}$	81.75%	82.89%
$\text{SRFeat}_{\text{IF}}$ [54]	88.07%	89.21%
E-DBPN$_{\text{G}}$	91.23%	92.11%
Baseline (HR)	95.26%	96.32%

In our work, we employ the pretrained CaffeNet network [87] obtained using the ImageNet data set [76]. Then, we extract the activations from the first fully connected layer and feed these activations into the support vector machines (SVMs) for classification. Note that the input images for the pretrained network should be resized according to the size of the receptive field. For the sake of thoroughly evaluating the effectiveness of various SISR methods in terms of the perceptual quality, we use two remote sensing scene data sets, i.e., the UC-Merced data set and the WHU-RS19 data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

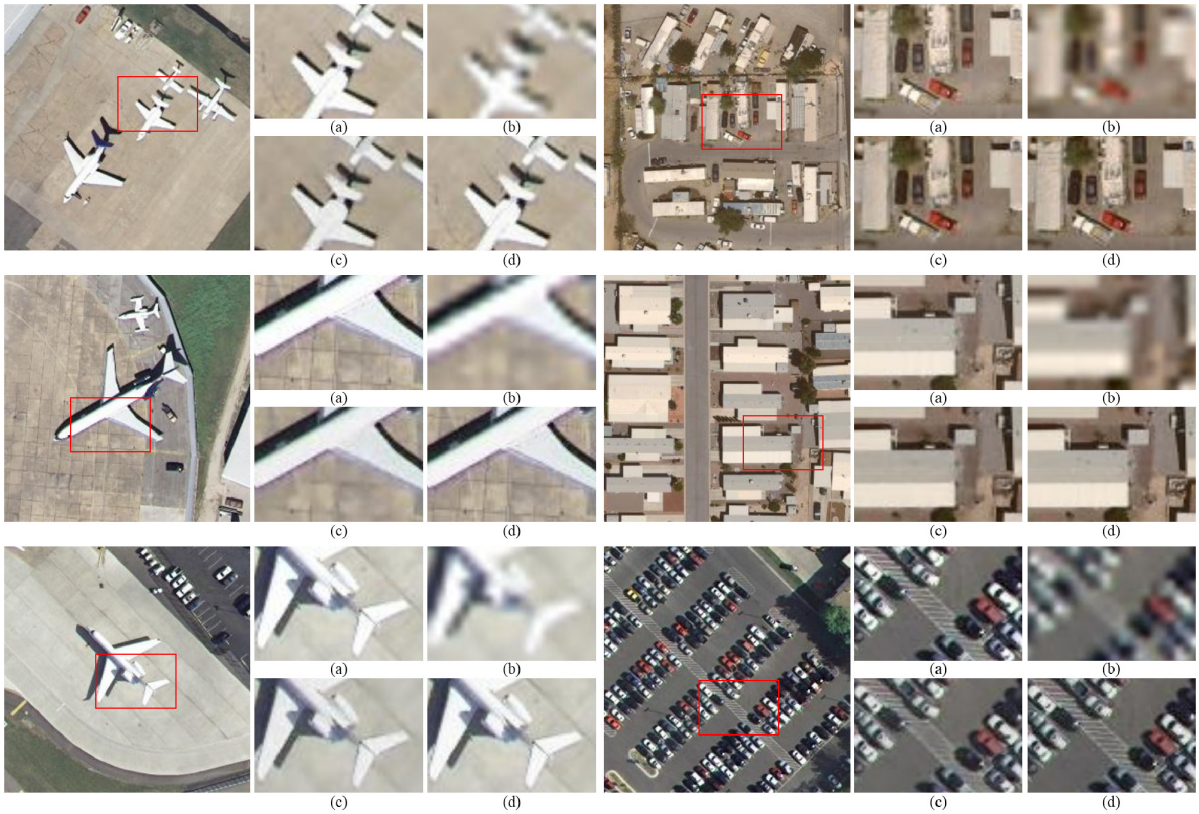


Fig. 7. Visual comparisons of the GAN-based SISR methods with our results on scale ×4. (a) HR. (b) Bicubic. (c) $\text{SRFeat}_{\text{IF}}$. (d) E-DBPN_{G} (ours).

set. At the same time, we adopt two different training-testing ratios for each utilized data set. For the UC-Merced data set, the ratio of training/testing images is set to 50:50 and 80:20, respectively. For the WHU-RS19 data set, we fix the ratio of the training set to be 40% and 60%, respectively, and the remaining images for testing. The activations of the training set are used to train the SVMs, and the activations of the testing set are used for calculating the classification accuracy. In order to compute the baseline score, we use the original HR images in the testing set directly. Next, we design four different degraded versions of the testing set by downsampling the original HR images and applying four different SISR methods: Bicubic, $\text{SRFeat}_{\text{IF}}$ [54], E-DBPN_{M}, and E-DBPN_{G}.

Tables VII and VIII show these classification accuracies on the UC-Merced data set and the WHU-RS19 data set, respectively. From the obtained results, we can see that our MSE-based E-DBPN, i.e., E-DBPN_{M}, does not perform well. The reason behind this is that the restored high-frequency details are not enough. As for our GAN-based model, i.e., E-DBPN_{G}, it achieves better performance, which is much closer to the baseline results compared with others. The good performance of E-DBPN_{G} indicates that the differences of the perceptual quality between the original HR images and our superresolved images are not significant. In summary, the results in Tables VII and VIII clearly demonstrate the effectiveness of our method.

We also make a qualitative comparison with GAN-based SISR methods, which is shown in Fig. 7. From the visual comparison in Fig. 7, we can see that the superresolved images produced by $\text{SRFeat}_{\text{IF}}$ have blurry details. Nevertheless, the superresolved images produced by our method have a better visual effect, which is very similar to the original HR images.

E. Evaluation on Real LR Remote Sensing Scene Images

In this section, we employ real LR remote sensing scene images to evaluate the robustness of our proposed method. These LR remote sensing scene images are from GaoFen-2 satellite imagery. However, in real-world SR scenes, we have only LR images, without corresponding HR images for reference. Hence, we should employ a nonreference image quality assessment method, i.e., average gradient (AG) [88]. Since the quantitative value of AG can sensitively capture content sharpness, detail contrast, and texture diversity from images, thus it is often employed for the assessment of image clarity. Generally, a larger AG indicates richer details.

We employ four different SISR methods, i.e., Bicubic, WTCRR [28], $\text{SRFeat}_{\text{IF}}$ [54], and E-DBPN_{G}, to conduct real LR remote sensing scene images SR. Fig. 8 shows the visual comparisons on real LR remote sensing scene images. From the visual comparisons in Fig. 8, we can observe that our proposed method still has a better visual effect compared with others. At the same time, we also give the corresponding quantitative value of AG below each image. Our proposed method achieves the higher AG value than other methods, which indicates that the images reconstructed by our proposed method, i.e., E-DBPN_{G}, have richer details.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

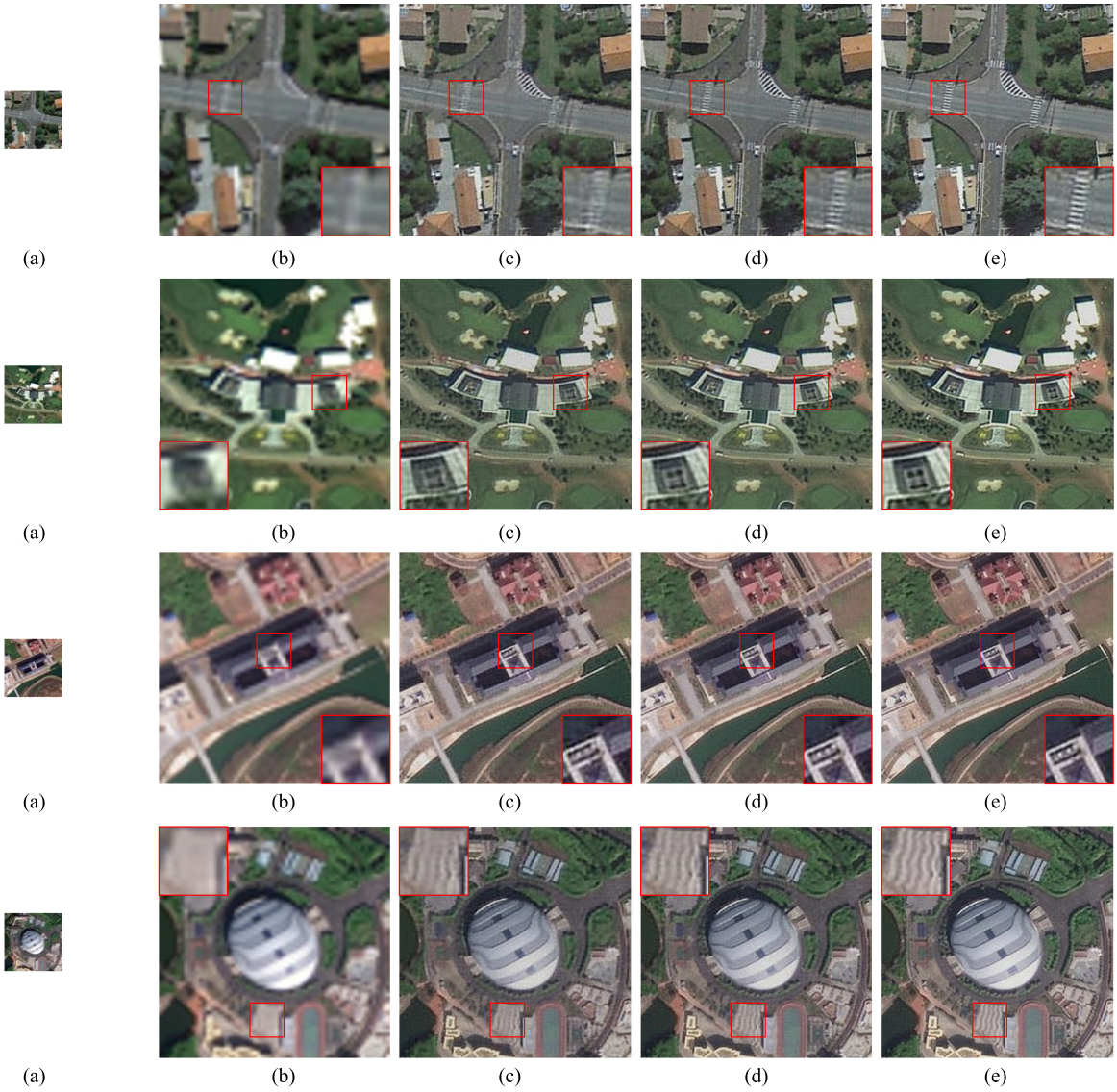


Fig. 8. Superresolution results of real LR remote sensing scene images on scale $\times 4$. (a) LR. (b) Bicubic. (c) WTCRR. (d) $\text{SRFeat}_{\text{IF}}$. (e) E-DBPN_{G} (ours).

V. Conclusion

In this article, we present a GAN-based SISR method by proposing a novel generator, which can generate perceptually pleasing remote sensing scene images. First, we design the E-DBPNs generator based on the architecture of the original DBPN and make two modifications for further improving the superresolved quality of DBPN. One is to add the proposed ERCAM into DBPN. The other is to replace the concatenation operation with the proposed SFFM for dealing with the feature maps generated by different up-projection units discriminatorily. As for the training, the E-DBPN generator is first trained based on the MSE loss. Next, the content loss and the adversarial loss are employed to train our initialized generator network for improving the perceptual quality of the recovered images. In the end, extensive experiments are conducted on five publicly available remote sensing scene data sets. The experimental results show that our proposed SISR method outperforms state-of-the-art SISR methods.

Inspired by the promising results produced from our method, in the future, we plan to extend our framework to other applications, e.g., hyperspectral image SR and multispectral image SR, etc.

References

[1] Y. Yuan, J. Fang, X. Lu, and Y. Feng, "Remote sensing image scene classification using rearranged local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1779–1792, Mar. 2019.

[2] B.-D. Liu, J. Meng, W.-Y. Xie, S. Shao, Y. Li, and Y. Wang, "Weighted spatial pyramid matching collaborative representation for remote-sensing-image scene classification," *Remote Sens.*, vol. 11, no. 5, p. 518, Mar. 2019.

[3] J. M. Oliver, "Current status of the use of small unmanned aerial systems for environmental monitoring," *Environ. Claims J.*, vol. 29, no. 2, pp. 159–170, Apr. 2017.

[4] A. Elfadaly, W. Attia, and R. Lasaponara, "Monitoring the environmental risks around medinet habu and Ramesseum temple at West Luxor, Egypt, using remote sensing and GIS techniques," *J. Archaeol. Method Theory*, vol. 25, no. 2, pp. 587–610, Jun. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

[5] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS J. Photogram. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.

[6] W. Xiong, Y. Lv, Y. Cui, X. Zhang, and X. Gu, "A discriminative feature learning approach for remote sensing image retrieval," *Remote Sens.*, vol. 11, no. 3, p. 281, Feb. 2019.

[7] W. Yang, X. Zhang, Y. Tian, W. Wang, and J.-H. Xue, "Deep learning for single image super-resolution: A brief review," Aug. 2018, *arXiv:1808.03344*. Accessed: Apr. 1, 2019. [Online]. Available: https://arxiv.org/abs/1808.03344

[8] V. K. Ha, J. Ren, X. Xu, S. Zhao, G. Xie, and V. M. Vargas, "Deep learning based single image super-resolution: A survey," in *Proc. Int. Conf. Brain Inspired Cognit. Syst.*, 2018, pp. 106–119.

[9] W. T. Freeman, T. R. Jones, and E. C. Pasztor, "Example-based super-resolution," *IEEE Comput. Graph. Appl.*, vol. 22, no. 2, pp. 56–65, Mar./Apr. 2002.

[10] H. Chang, D.-Y. Yeung, and Y. Xiong, "Super-resolution through neighbor embedding," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun./Jul. 2004, p. 1.

[11] D. Glasner, S. Bagon, and M. Irani, "Super-resolution from a single image," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 349–356.

[12] G. Freedman and R. Fattal, "Image and video upscaling from local self-examples," *TOGACM Trans. Graph.*, vol. 30, no. 2, pp. 1–11, Apr. 2011.

[13] C. E. Duchon, "Lanczos filtering in one and two dimensions," *J. Appl. Meteor.*, vol. 18, no. 8, pp. 1016–1022, Aug. 1979.

[14] R. Keys, "Cubic convolution interpolation for digital image processing," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-29, no. 6, pp. 1153–1160, Dec. 1981.

[15] M. Irani and S. Peleg, "Improving resolution by image registration," *CVGIP, Graph. Models Image Process.*, vol. 53, no. 3, pp. 231–239, May 1991.

[16] R. Schultz and R. Stevenson, "Extraction of high-resolution frames from video sequences," *IEEE Trans. Image Process.*, vol. 5, no. 6, pp. 996–1011, Jun. 1996.

[17] N. Nguyen, P. Milanfar, and G. Golub, "Efficient generalized cross-validation with applications to parametric image restoration and resolution enhancement," *IEEE Trans. Image Process.*, vol. 10, no. 9, pp. 1299–1308, 2001.

[18] J. Sun, Z. Xu, and H.-Y. Shum, "Image super-resolution using gradient profile prior," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008, pp. 1–8.

[19] J. Yang, J. Wright, T. Huang, and Y. Ma, "Image super-resolution as sparse representation of raw image patches," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008, pp. 1–8.

[20] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[21] B. Hou, K. Zhou, and L. Jiao, "Adaptive super-resolution for remote sensing images based on sparse representation with global joint dictionary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2312–2327, Apr. 2018.

[22] S. Bao and A. C. S. Chung, "Multi-scale structured CNN with label consistency for brain MR image segmentation," *Comput. Methods Biomech. Biomed. Eng., Imag. Visualizat.*, vol. 6, no. 1, pp. 113–117, Jan. 2018.

[23] J. Dolz, K. Gopinath, J. Yuan, H. Lombaert, C. Desrosiers, and I. B. Ayed, "HyperDense-Net: A hyper-densely connected CNN for multi-modal image segmentation," *IEEE Trans. Med. Imaging*, vol. 38, no. 5, pp. 1116–1126, May 2019.

[24] D. Han, Q. Liu, and W. Fan, "A new image classification method using CNN transfer learning and Web data augmentation," *Expert Syst. Appl.*, vol. 95, pp. 43–56, Apr. 2018.

[25] L. Song *et al.*, "A deep multi-modal CNN for multi-instance multi-label image classification," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 6025–6038, Dec. 2018.

[26] F. Radenovic, G. Tolias, and O. Chum, "Fine-tuning CNN image retrieval with no human annotation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 7, pp. 1655–1668, Jul. 2019.

[27] A. Khatami, M. Babaie, H. Tizhoosh, A. Khosravi, T. Nguyen, and S. Nahavandi, "A sequential search-space shrinking using CNN transfer learning and a Radon projection pool for medical image retrieval," *Expert Syst. Appl.*, vol. 100, pp. 224–233, Jun. 2018.

[28] W. Ma, Z. Pan, J. Guo, and B. Lei, "Achieving super-resolution remote sensing images via the wavelet transform combined with the recursive res-net," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3512–3527, Jun. 2019.

[29] S. Lei, Z. Shi, and Z. Zou, "Super-resolution for remote sensing images via local–global combined network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1243–1247, Aug. 2017.

[30] Y. Luo, L. Zhou, S. Wang, and Z. Wang, "Video satellite imagery super resolution via convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2398–2402, Dec. 2017.

[31] K. Jiang, Z. Wang, P. Yi, J. Jiang, J. Xiao, and Y. Yao, "Deep distillation recursive network for remote sensing imagery super-resolution," *Remote Sens.*, vol. 10, no. 11, p. 1700, Oct. 2018.

[32] T. Wang, W. Sun, H. Qi, and P. Ren, "Aerial image super resolution via wavelet multiscale convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 769–773, May 2018.

[33] A. Xiao, Z. Wang, L. Wang, and Y. Ren, "Super-resolution for 'Jilin-1' satellite video imagery via a convolutional network," *Sensors*, vol. 18, no. 4, p. 1194, Apr. 2018.

[34] K. Jiang, Z. Wang, P. Yi, and J. Jiang, "A progressively enhanced network for video satellite imagery superresolution," *IEEE Signal Process. Lett.*, vol. 25, no. 11, pp. 1630–1634, Nov. 2018.

[35] M. Haris, G. Shakhnarovich, and N. Ukita, "Deep back-projection networks for super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1664–1673.

[36] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 184–199.

[37] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[38] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[39] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1637–1645.

[40] Y. Tai, J. Yang, X. Liu, and C. Xu, "MemNet: A persistent memory network for image restoration," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4539–4547.

[41] Y. Tai, J. Yang, and X. Liu, "Image super-resolution via deep recursive residual network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3147–3155.

[42] A. Shocher, N. Cohen, and M. Irani, "Zero-shot super-resolution using deep internal learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3118–3126.

[43] P. Ren, W. Sun, C. Luo, and A. Hussain, "Clustering-oriented multiple convolutional neural networks for single image super-resolution," *Cogn. Comput.*, vol. 10, no. 1, pp. 165–178, Feb. 2018.

[44] Z. Zhang, X. Wang, and C. Jung, "DCSR: Dilated convolutions for single image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1625–1635, Apr. 2019.

[45] W. Shi *et al.*, "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1874–1883.

[46] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 391–407.

[47] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4681–4690.

[48] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.

[49] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4799–4807.

[50] M. S. M. Sajjadi, B. Scholkopf, and M. Hirsch, "EnhanceNet: Single image super-resolution through automated texture synthesis," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4491–4500.

[51] W. Han, S. Chang, D. Liu, M. Yu, M. Witbrock, and T. S. Huang, "Image super-resolution via dual-state recurrent networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1654–1663.

[52] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.

[53] X. Wang *et al.*, "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2019, pp. 63–79.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.

[54] S.-J. Park, H. Son, S. Cho, K.-S. Hong, and S. Lee, "SRFeat: Single image super-resolution with feature discrimination," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 455–471.
[55] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 624–632.
[56] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Fast and accurate image super-resolution with deep Laplacian pyramid networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 11, pp. 2599–2613, Nov. 2019.
[57] Y. Wang, F. Perazzi, B. Mcwilliams, A. Sorkine-Hornung, O. Sorkine-Hornung, and C. Schroers, "A fully progressive approach to single-image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 864–873.
[58] Z. Wang, J. Chen, and S. C. H. Hoi, "Deep learning for image super-resolution: A survey," Feb. 2019, *arXiv:1902.06068*. Accessed: Apr. 1, 2019. [Online]. Available: https://arxiv.org/abs/1902.06068
[59] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[60] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, vol. 2, Jul. 2004, pp. 1398–1402.
[61] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," Sep. 2014, *arXiv:1409.1556*. Accessed: Apr. 1, 2019. [Online]. Available: https://arxiv.org/abs/1409.1556
[62] A. Dosovitskiy and T. Brox, "Generating images with perceptual similarity metrics based on deep networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 658–666.
[63] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 694–711.
[64] X. Wang, K. Yu, C. Dong, and C. C. Loy, "Recovering realistic texture in image super-resolution by deep spatial feature transform," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 606–615.
[65] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
[66] L. Gatys, A. S. Ecker, and M. Bethge, "Texture synthesis using convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 262–270.
[67] L. A. Gatys, A. S. Ecker, and M. Bethge, "Image style transfer using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2414–2423.
[68] Y. Yuan, S. Liu, J. Zhang, Y. Zhang, C. Dong, and L. Lin, "Unsupervised image super-resolution using cycle-in-cycle generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 701–710.
[69] H. Aly and E. Dubois, "Image up-sampling using total-variation regularization with a new observation model," *IEEE Trans. Image Process.*, vol. 14, no. 10, pp. 1647–1659, Oct. 2005.
[70] A. Bulat and G. Tzimiropoulos, "Super-FAN: Integrated facial landmark localization and super-resolution of real-world low resolution faces in arbitrary poses with GANs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 109–117.
[71] M. Corbetta and G. L. Shulman, "Control of goal-directed and stimulus-driven attention in the brain," *Nature Rev. Neurosci.*, vol. 3, no. 3, pp. 201–215, Mar. 2002.
[72] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
[73] J. Park, S. Woo, J.-Y. Lee, and I. S. Kweon, "BAM: Bottleneck attention module," Jul. 2018, *arXiv:1807.06514*. Accessed: Apr. 1, 2019. [Online]. Available: https://arxiv.org/abs/1807.06514
[74] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[75] W. Rawat and Z. Wang, "Deep convolutional neural networks for image classification: A comprehensive review," *Neural Comput.*, vol. 29, no. 9, pp. 2352–2449, Sep. 2017.
[76] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.
[77] E. Agustsson and R. Timofte, "NTIRE 2017 challenge on single image super-resolution: Dataset and study," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 126–135.
[78] M. Bevilacqua, A. Roumy, C. Guillemot, and M.-L.-A. Morel, "Low-complexity single-image super-resolution based on nonnegative neighbor embedding," in *Proc. Brit. Mach. Vis. Conf.*, 2012.
[79] R. Zeyde, M. Elad, and M. Protter, "On single image scale-up using sparse-representations," in *Proc. Int. Conf. Curves Surfaces*, 2010, pp. 711–730.
[80] D. Martin, C. Fowlkes, D. Tal, and J. Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," in *Proc. 8th IEEE Int. Conf. Comput. Vis. (ICCV)*, Vancouver, BC, Canada, Nov. 2002, pp. 416–423.
[81] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst. (GIS)*. New York, NY, USA: ACM, 2010, pp. 270–279.
[82] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, Apr. 2012.
[83] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.
[84] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[85] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[86] C. Fookes, F. Lin, V. Chandran, and S. Sridharan, "Evaluation of image resolution and super-resolution on face recognition performance," *J. Vis. Commun. Image Represent.*, vol. 23, no. 1, pp. 75–93, Jan. 2012.
[87] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. 22nd ACM Int. Conf. Multimedia*. New York, NY, USA: ACM, 2014, pp. 675–678.
[88] A. A. Chen, X. Chai, B. Chen, R. Bian, and Q. Chen, "A novel stochastic stratified average gradient method: Convergence rate and its complexity," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2018.



Yunlong Yu received the B.S. and M.S. degrees from the Air Force Engineering University, Xi'an, China, in 2014 and 2017, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include remote sensing, image classification, and deep learning.



Xianzhi Li (Student Member, IEEE) received the B.S. degree from the Sichuan University, Chengdu, China, in 2014, and the M.Sc. degree from the Chinese University of Hong Kong, Hong Kong, in 2015, where she is currently pursuing the Ph.D. degree with the Department of Computer Science and Engineering.

Her research interests include computer graphics, geometry processing, and deep learning.



Fuxian Liu received the B.S. degree from the Lanzhou University, Lanzhou, China, in 1994, and the M.S. and Ph.D. degrees from the Air Force Engineering University, Xi'an, China, in 1998 and 2001, respectively.

He is currently a Professor with the Air Force Engineering University. His research interests include deep learning and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:13:32 UTC from IEEE Xplore. Restrictions apply.